# Function & performance

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe how function influences the design of a product
- ☐ explain primary and secondary functions with examples
- ☐ describe how performance and fitness for purpose influence design
- ☐ describe maintenance issues associated with products
- ☐ explain the influence of planned obsolescence on manufacturers, consumers and the environment
- ☐ explain value for money

## I DO — Worked example

Read how a full-mark answer is built

*A folding camping chair and a fixed garden recliner.*

**1 Describe how function and safety have influenced the design of these chairs.** (5 marks)

**Points that earn marks** from the marking instructions: one ✓ per point

- ✓ Function: camping chair folds flat to fit in a car boot
- ✓ Function: cup holder in armrest keeps drinks within reach
- ✓ Function: recliner back adjusts to several positions
- ✓ Safety: wide leg spread keeps the chair stable on uneven ground
- ✓ Safety: locking catch stops the frame folding while in use

**Why marks are not awarded**

- ✗ Naming a function with no description
- ✗ Only one factor covered: max 4

## WE DO — Guided practice

Use the starters and word bank to help you

**2 Describe planned obsolescence and give an example of its influence on the design of products.** (2 marks)

**Sentence starters**

Description: …
Example: …

**Word bank**

outdated · limited · cannot · become · description · replaced · designed · batteries

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** A razor with replaceable heads compared with a disposable razor has...
☐ Shorter life expectancy ☐ Longer life expectancy ☐ No maintenance ☐ No secondary function

**b)** Making a phone battery hard to replace mainly affects which factor?
☐ Aesthetics ☐ Maintenance ☐ Target market ☐ Anthropometrics

**c)** A torch that also works as a bottle opener. The bottle opener is...
☐ Primary function ☐ Secondary function ☐ Planned obsolescence ☐ Aesthetics

**d)** Which is a benefit of planned obsolescence for a manufacturer?
☐ Fewer sales ☐ Predictable repeat sales ☐ No need for R&D ☐ Better reputation guaranteed

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Primary function** | ____ | **A** Who maintains it, what tasks are needed, and how easily parts can be accessed or replaced. |
| 2. **Secondary function** | ____ | **B** Designing products with a limited life so they will be replaced. |
| 3. **Fitness for purpose** | ____ | **C** Additional jobs that add value to the product. |
| 4. **Maintenance consideration** | ____ | **D** How well a product meets its intended use, user, environment and life expectancy. |
| 5. **Planned obsolescence** | ____ | **E** The main job a product is designed to do. |

### 5 Explain what is meant by ‘Style obsolescence’. (2 marks)

### 6 Explain what is meant by ‘Value for money’. (2 marks)

### Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?